# Tiered Neurodata Risk Classification Framework based on functional exposure

*Classifying neurodata by the capacity to enable inference, reconstruction, profiling or harm.*

| RISK CATEGORY | RISK TIER & DESCRIPTION | | ILLUSTRATIVE ARTEFACTS REQUIRING ASSESSMENT |
|---|---|---|---|
| **Conventional privacy-based risk**<br>TIERS 1, 2 AND 3<br>Risk linked primarily to aggregation, participant linkage, identifiability and traceability. | 1 | **TIER 1 — AGGREGATED OR FUNCTIONALLY NON-REVEALING NEURODATA**<br>Aggregated, statistical or summary-level information that contains no participant-level record, persistent participant trace or linkage mechanism and presents no reasonable small-group disclosure risk. | • Group-average brain-activation patterns<br>• Aggregated connectivity measures and summary statistics<br>• Published statistical results with no participant-level records<br>• Sufficiently aggregated outputs from which no individual or small group can reasonably be identified |
| | 2 | **TIER 2 — PSEUDONYMISED OR PARTICIPANT-LINKABLE NEURODATA**<br>Participant-level neurodata from which direct identifiers have been removed but which may still be linked to an individual through a code, repository record, institutional key or other retained linkage mechanism. | • Pseudonymised EEG or MEG data linked to a participant code<br>• De-identified MRI capable of re-linkage through a secure institutional key<br>• Participant-level processed neurodata held in a controlled repository<br>• Coded behavioural-neural datasets |
| **Post-transformation functional-exposure tiers** | 3 | **TIER 3 — DIRECTLY IDENTIFIABLE OR INTRINSICALLY TRACEABLE NEURODATA**<br>Neurodata containing direct identifiers, recognisable identifying anatomy, identifying acquisition metadata or intrinsic signal characteristics that reasonably permit identification or tracing without relying solely on a separately protected code-to-identity mapping. A raw acquisition format does not, by itself, establish Tier 3. | • Non-defaced structural imaging; raw DICOM with headers or recognisable anatomy<br>• EEG/MEG/ECoG retaining identifying metadata or demonstrated participant-tracing capacity<br>• Implant telemetry or BCI recordings with device identifiers or precise timestamps<br>• Neurodata associated with names, voice, video or other direct identifiers |

TIERS 4 AND 5

Risk driven by the capacity to support inference, profiling or harm, independent of identifiability.

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| 4 | **TIER 4 — FUNCTIONALLY REVEALING DERIVATIVE NEURODATA**<br>Transformed or derived neurodata that may no longer resemble the source dataset but can reasonably reveal, predict, reconstruct or profile neurological, cognitive, emotional or behavioural states. | • Neural feature vectors, embeddings and neural fingerprints<br>• Cognitive or behavioural profiles; classifiers and predictive outputs<br>• Reconstructed visual, auditory or cognitive representations<br>• Synthetic neurodata or latent representations retaining meaningful neural characteristics |
| 5 | **TIER 5 — MODEL-ASSOCIATED OR COMPUTATIONALLY PERSISTENT NEURODATA**<br>Neurodata-derived information associated with an AI model or persistent computational infrastructure where the artefact retains, or may reasonably retain, continuing functional exposure through inference, reconstruction, extraction, profiling or memorisation. Incorporation into a model triggers assessment; it does not by itself establish Tier 5. | • Model weights trained on neurodata; neuroscience foundation models<br>• Federated-learning systems and synthetic-data generators trained on neurodata<br>• Persistent embedding repositories<br>• AI systems that retain neurodata-derived information after the original research activity |

**KEY PRINCIPLE** Tier assignment is informed by functional exposure; legal status remains determined under applicable law. The highest applicable tier governs; safeguards are cumulative; an Unknown answer is governed under the more protective tier until resolved; incorporation into a model triggers assessment but does not by itself establish Tier 5.